

transpose: interchange rows and columns $\Rightarrow A = \begin{bmatrix} 0 & 8 \\ 1 & 3 \end{bmatrix}$ $A^t = \begin{bmatrix} 0 & 1 \\ 8 & 3 \end{bmatrix}$

THEOREM 2.4

Properties of Transpose

Let A and B be matrices and c be a scalar. Assume that the sizes of the matrices are such that the operations can be performed.

1. $(A + B)^t = A^t + B^t$ Transpose of a sum
2. $(cA)^t = cA^t$ Transpose of a scalar multiple
3. $(AB)^t = B^t A^t$ Transpose of a product
4. $(A^t)^t = A$

Symmetric matrix: A matrix that is equal to its transpose $A = A^t$

Trace of matrix: Sum of diagonal elements of a square matrix.

THEOREM 2.5

Properties of Trace

Let A and B be matrices and c be a scalar. Assume that the sizes of the matrices are such that the operations can be performed.

1. $tr(A + B) = tr(A) + tr(B)$
2. $tr(AB) = tr(BA)$
3. $tr(cA) = ctr(A)$
4. $tr(A^t) = tr(A)$

Ch. 2.4

Inverse of matrix:

DEFINITION

Let A be an $n \times n$ matrix. If a matrix B can be found such that $AB = BA = I_n$, then A is said to be invertible and B is called an inverse of A . If such a matrix B does not exist, then A has no inverse.

$$AA^{-1} = I = A^{-1}A$$

Unique: If a matrix is invertible, that inverse is unique.

Properties of Matrix Inverse

Let A and B be invertible matrices and c a nonzero scalar. Then

1. $(A^{-1})^{-1} = A$
2. $(cA)^{-1} = \frac{1}{c}A^{-1}$
3. $(AB)^{-1} = B^{-1}A^{-1}$
4. $(A^n)^{-1} = (A^{-1})^n$
5. $(A^t)^{-1} = (A^{-1})^t$

$A_{n \times n} X = Y$. if A^{-1} exists $\Rightarrow A$ is row equivalent to $I_n \Rightarrow$ unique solution.
and the solution: $X = A^{-1}Y$

Ch. 2.2

Matrix of linear combination: $\begin{bmatrix} a_{11}x_1 & \dots & a_{1n}x_n \\ \vdots & & \vdots \\ a_{m1}x_1 & \dots & a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \Rightarrow \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$
 $\Rightarrow AX = B$

Subspace: $A_1X=0, A_2X=0 \Rightarrow (A_1+A_2)X=0$ closed under addition

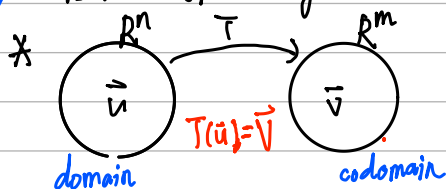
$cA_1X=0 \Rightarrow A_1(cX)=0$ closed under scalar multiplication

* The sets of solutions to a homogeneous system of linear equation is a subspace
but to a non homogeneous system $AX=B$: $A_1X=B \Rightarrow (A_1+A_2)X=2B$
 $A_2X=B$

* The sets of solutions to a non homogeneous system of linear equation is not a subspace

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Transformation Mapping T : A rule that assigns to each $\vec{u}$ in R^n a unique $\vec{v}$ in R^m . R^n is domain of T . R^m is codomain of T .

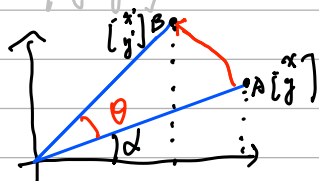


$\vec{v}$ is image of $\vec{u}$ under T

Dilation: $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow r > 1$

Contraction: $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow 0 < r < 1$

Reflection: $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$

Rotation:  $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$ (rotation about the origin)